So, a is not unique any no. greater than a will be upper bound of set S.
Further, such $a \in \mathbb{R}$ is called an upper bound of S

Axiom (Least upper bound property):
Any non empty set of real numbers with an upper bound has a least upper bound.

i) If $S \subset \mathbb{R}$ is non-empty and $\exists\ a \in \mathbb{R}$ $x \le a\ \forall\ x \in S$.

ii) then there is a number $a_0 \in \mathbb{R}$ such that

i) a_0 is an upper bound of S. ii) If b is any upper bound of S, then $a_0 \le b$.

Notation:- a_0 = least upper bound S.
$a_0 = (l.u.b)S$ → (Doubt)
$a_0 = \sup S$

(*) Similarly one can define the concepts of lower bound and greatest lower bound.

11/3/25

Lec-2

Property (l.u.b):-

$S \subset \mathbb{R}$ and bounded above.

$\Rightarrow \exists\ a_0 \to$ l.u.b of S.

Similarly, we can get g. lower bound of S.

Theorem:- If x and y are two positive numbers. Then $\exists\ n \in \mathbb{N}$ s.t $\boxed{nx > y}$ (Archamedian property)

Proof:- We prove it by contradiction

Suppose $nx \leq y \quad \forall n \in \mathbb{N}$.

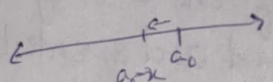
Then $A = \{nx : n \in \mathbb{N}\}$

Then A is bounded by above. Then by least upper bound property $\exists\ a_0 \in \mathbb{R}$ s.t.

$\sup A = a_0$

By the definition of Supremum $a_0 - x \leq n_0 x \leq a_0$ for some $n_0 \in \mathbb{N}$

$\Leftrightarrow a_0 \leq (n_0+1)x$ (Contradiction)



Then a_0 cannot be an upper bound of A.

So, $\therefore \boxed{nx > y}$ proved $\Rightarrow$

Theorem:- If $a, b \in \mathbb{R}$ such that $a < b$ then there exist a rational number $r \in \mathbb{Q}$ s.t

$\boxed{a \leq r < b}$.

Proof Given $b - a > 0$

In previous theorem, put $x = b-a$ and $y = 1$

Then $\exists\ n_0 \in \mathbb{N}$ such that (Converse)

$n_0(b-a) > 1$ (Distance btw any 2 natural numbers is 1 but here it's more than 1 so, there must be a rational number in btw)

Therefore, $\exists\ p \in \mathbb{Z}$ such that $n_0 a \leq p < n_0 b$

Then $a \leq \frac{p}{n_0} < b$ where $\frac{p}{n_0}$ is rational number.

Theorem:- A monotone, bounded sequence of real numbers converges.

Proof: Let $\{a_n\}$ be the bounded non-decreasing monotone sequence (set).

Sequence:- $f : \mathbb{N} \to \mathbb{R}$

function of natural numbers to real numbers.

$\Rightarrow$ Consider $a_0 = \sup a_n$

We want to show that $a_0 = \lim_{n \to \infty} a_n$

Let $\varepsilon > 0$, then by the defination

$a_0 - \varepsilon < a_n < a_0$ for some $n \in \mathbb{N}$

As $\{a_n\}$ is non-decreasing then.

$a_0 - \varepsilon < a_N \leq a_{N+1} \leq \cdots \leq a_0$

$a_0 - \varepsilon < a_n < a_0 \quad \forall\ n \geq N$

$\Rightarrow |a_0 - a_n| < \varepsilon \quad \forall\ n \geq N$

Convergence defination.

Here $a_0 = \lim_{n \to \infty} a_n$

If $\{a_n\}$ is monotone non-increasing then we take $b_n = -a_n$ and use the above result.

Rec-3 4/3/25

Property (Nested interval Property):
If I_n is a sequence of closed bounded non-empty intervals of $\mathbb{R}$ such that
$I_1 \supset I_2 \supset I_3 \supset \cdots \supset I_{n-1} \supset I_n$
Then $\bigcap_{n=1}^{\infty} I_n \neq \phi$

If in addition if length $(I_n) \to 0$ then $\bigcap_{n=1}^{\infty} I_n$ = {a} it contains exactly one point.

eg: i) $I_n = [n, \infty)$ $I_n \cap I_n = \phi$
ii) $I_n = (0, \frac{1}{n}]$ $\bigcap_{n=1}^{\infty} I_n = \phi$



Recall:
Theorm: {I_n} intervals → closed → bounded and $I_n \supset I_{n+1}$ ...
Then $\bigcap_{n=1}^{\infty} I_n$ is non empty.
If addition; length$(I_n) \to 0$
Then $\bigcap_{n=1}^{\infty} I_n = \{x\}$.

Proof: – Let $I_n = [a_n, b_n]$ $\forall n \in \mathbb{N}$
Then as $I_n \supset I_{n+1}$
$a_n \leq a_{n+1} \cdots \leq b_{n+1} < b_n < \cdots < b_1$
$\{a_n\}$ is non-decreasing seq and $\{b_n\}$ is non-increasing. (bounded)
$\lim a_n = \sup a_n = a$
and $\lim b_n = \inf b_n = b$

Claim: – $\bigcap_{n=1}^{\infty} I_n = [a, b]$.

To see, $x \in \bigcap_{n=1}^{\infty} I_n \Rightarrow x \in I_n \; \forall n$

So, $a_n \leq x \leq b_n \quad \forall n \in \mathbb{N}$.
As $a = \sup a_n$, we find $a \leq x$
Similarly $b = \inf b_n$ we find $x \leq b$.
$\Rightarrow x \in [a, b]$
$\bigcap_{i=1}^{\infty} I_n \subset [a, b]$

Next; $y \in [a, b]$.
$a_n \leq a \leq y \leq b \leq b_n \quad \forall n \in \mathbb{N}$
Hence $y \in \bigcap_{n=1}^{\infty} [a_n, b_n] = I_n$ proved.

Hint = length $(I_n) = b_n - a_n$

If $b_n - a_n \xrightarrow[n\to\infty]{} 0$, we show $\bigcap_{n=1}^{\infty} I_n = \{x\} = \{b\}$

They will be the same point. (Proved)

Remark :- Convergence of monotone bounded seq / Nested interval Theorem is equivalent to assuming l.u.b property.

(or) Taking one as an axiom we can prove the other properties.

Limit and Continuity :-

Assume :- we all know the defn of limit and Continuity.

⊙ Limit supremum / Limit infimum

1) If you have a seq $\{a_n\}$ bounded + monotone ⇒ a_n is convergent.

Eg $a_n = \begin{cases} -1, & n \text{ odd} \\ 1, & n = \text{even} \end{cases}$

it is not a convergent seq, it is an oscillating seq btw $[-1, 1]$.

Defn: Let $\{a_n\}$ be a seq

$\{T_n\} = \sup_n \{a_k : k \geq n\}$ (it is a subset of $\{a_n\}$)

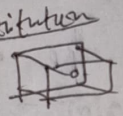
$\{t_n\} = \inf_n \{a_k : k \geq n\}$ (it is not always exist)

$t_n \nearrow T_n \in [-\infty, +\infty]$ (so if it does not exist sup and inf then sup = $+\infty$ inf = $-\infty$)

↙ Closed brackets

if open bracket it will be Real numbers system.

Eg (Institution)



particle in a box with ∞ particles.

Can they can be far apart no → they can be close to each other as well.

we define

$\lim_{n\to\infty} \sup a_n = \inf_n T_n = \lim_{n\to\infty} a_n$

$\liminf_{n\to\infty} a_n = \sup_n t_n = \lim_{n\to\infty} a_n$

⊙ $S_n = \{a_n : n \geq K\}$

bigger set → superset

$S_n \supseteq S_{n+1}$

Therefore $T_n \geq T_{n+1}$

$\{T_n\}$ is monotonically non-increasing.

Then $\lim_{n\to\infty} T_n = \inf_n T_n$

(assume $\{a_n\}$ is bounded)

similarly ⊙ $\{t_n\}$ is non-decreasing

(or) then $\lim_{n\to\infty} t_n = \sup_n t_n$

Eg i) $a_n = \frac{1}{n}$ (This is convergence)

$\limsup_{n\to\infty} a_n$?

$T_n = \sup_n \{\frac{1}{k} : k \geq n\} = \frac{1}{n}$

↙ greatest of all the no.s in the set

$\inf T_n = 0$

$\limsup_{n\to\infty} a_n = 0$

$\liminf_{n\to\infty} a_n$ = ?

$t_n = \inf \{\frac{1}{k} : k \geq n\} = 0$

$\liminf_{n\to\infty} a_n = 0$

Observation: for $a_n = \frac{1}{n}$, convergent seq

$\limsup_{n\to\infty} a_n = \liminf_{n\to\infty} a_n$

Q) For any convergent seq whether $\limsup_{n\to\infty} a_n = \liminf_{n\to\infty} a_n$?

Ans yes

Q) given bounded seq $\{a_n\}$. if $\limsup_{n\to\infty} a_n = \liminf_{n\to\infty} a_n$ whether $\{a_n\}$ Convergent??

Ans yes !!

Eg ii) $a_n = (-1)^n + \frac{1}{n}$, $n \in \mathbb{N}$ (bounded seq $a_n = [-2, +2)$)

To calculate $\overline{\lim}$ and $\underline{\lim}$

→ We find $T_n = \sup\left\{(-1)^k + \frac{1}{k}, k \geq n\right\}$

$= 1 + \frac{1}{n}$ / (or $1 + \frac{1}{n+1}$)

k → is even or odd)

$\overline{\lim_{n\to\infty}} a_n = \inf T_n = 1$

$t_n = \inf_n \left\{(-1)^k + \frac{1}{k} \cdot k \geq n\right\} = -1$

$= -1 + \frac{1}{n} / -1 + \frac{1}{n+1}$

$\underline{\lim_{n\to\infty}} a_n = \sup t_n = -1$ ~~$\neq \overline{\lim} a_n$~~

$\overline{\lim} a_n = \underline{\lim} a_n$

Remark:- If $\{a_n\}$ is bounded; for every $\varepsilon > 0$ ($\mu = \overline{\lim} a_n$ ↳ finite no.)

Then for every $\varepsilon > 0$. but it's bounded we have $a_n < \mu + \varepsilon$ for all but finitely many points.

and $\mu - \varepsilon < a_n$ for infinitely many points.



Lec-4

$\limsup a_n = \overline{\lim} a_n = \inf T_n$

where $T_n = \sup\{a_k : k \geq n\}$.

$\liminf a_n = \underline{\lim} a_n = \sup t_n$

$t_n = \inf\{a_k : k \geq n\}$

(a) Assume $\mu = \overline{\lim} a_n$, then for any $\varepsilon > 0$.

$a_n \leq \mu + \varepsilon \quad \forall n \geq N_0$

and $\mu - \varepsilon \leq a_n$ for infinitely many.

Eg $a_n = (-1)^n + \frac{1}{n}$

$\mu = \overline{\lim} a_n = 1$ Then, for any $\varepsilon > 0$

$a_n < 1 + \varepsilon$ for $\forall n \geq n_0$

$1 - \varepsilon < a_n$ for n even

Theorem: Every bounded sequence of real numbers has a convergent subsequences.

Proof Let $\mu = \overline{\lim} a_n < \infty$ we aim to show that $\exists$ a subseq $\{a_{n_k}\}$ s.t

$\mu = \lim_{k \to \infty} a_{n_k}$

$\mu = \inf_n T_n$. Choose $\varepsilon = 1$

then $\exists\, n_0$ s.t $\mu \leq T_{n_0} < \mu + \frac{1}{2}\left(\frac{\varepsilon}{2}\right)$

Now, $T_{n_0} = \sup\{a_k : k \geq n_0\}$

then

$T_{n_0} - \frac{1}{2} < \bar{a}_1$ where $\bar{a}_1 \in \{a_k : k \geq n_0\}$

$\therefore |\mu - \bar{a}_1| \leq |\mu - T_{n_0}| + |T_{n_0} - \bar{a}_1| <$

$\frac{1}{2} + \frac{1}{2} = 1 = \varepsilon_1$

$\therefore \boxed{|\mu - \bar{a}_1| < 1}$

Now we assume $\varepsilon_2 = \frac{1}{2}$

By defn of M. we find $\exists\, n_1$ s.t

$T_{n_1} < \mu + \frac{1}{4}$

As $T_{n_1} = \sup\{a_k : k \geq n_1\}$

$T_n - \frac{1}{4} < \bar{a}_2$ where $\bar{a}_2 \in \{a_k : k \geq n_0\}$

Therefore,

$|\mu - \bar{a}_2| < \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$

We find that $\exists\, \bar{a}_n \in \{a_n\}$.

s.t $|\mu - \bar{a}_n| < \frac{1}{n} \quad \forall n \in \mathbb{N}$

Choose the subseq as $\{\bar{a}_n\} \subset \{a_n\}$

It is clear that.

$\lim_{n \to \infty} \bar{a}_n = \mu$ (Bound)

Proposition:- If $\{a_n\} \to$ a convergent seq then $\liminf a_n = \underline{\lim} a_n = \lim a_n$

Theorem:- If $\{a_n\}$ is bounded and if $\underline{\lim} a_n = \overline{\lim} a_n$

Then $\{a_n\} \to$ convergent.

Proof:- $a = \overline{\lim} a_n = \underline{\lim} a_n$ let $\varepsilon > 0$ then by the defn of limsup we see

$a_n < a + \varepsilon \quad \forall n \geq N_1$

Again by the defn of lim-inf

$a - \varepsilon < a_n \quad \forall n \geq N_2$

Choose $N = \max\{N_1, N_2\}$ then we get $|a - a_n| < \varepsilon \quad \forall n \geq N$

$= \lim_{n\to\infty} a_n = a$

Theorem: A seq of real numbers converges iff $\{a_n\}$ is Cauchy

Proof: Assume $\{a_n\}$ is Cauchy.

Let $\varepsilon > 0$, then $\exists\ N \in \mathbb{N}$ s.t

$|a_n - a_m| < \varepsilon \quad \forall n, m \geq N$

In particular

$a_N - \varepsilon < a_n < a_n + \varepsilon \quad , \forall n \geq N$

Therefore, a_n is a bounded seq.

Then $\limsup a_n$ and $\liminf a_n$ are finite

We also get $a_N - \varepsilon < \liminf a_n$

[Note that $a_N - \varepsilon \leq t_n \quad \forall k \geq N$

$a_N - \varepsilon \leq \sup t_k = \liminf_n$]

Similarly, we can also find that

$\limsup a_n < a_N + \varepsilon$

Then,

$\limsup a_n - \liminf a_n \leq a_N + \varepsilon - a_N + \varepsilon = 2\varepsilon$

as ε is arbitary no. so it can be as close as we can think of.

We get that.

$\limsup a_n = \liminf a_n$

Then $\{a_n\}$ is Convergent (proved)

Cauchy seq:

$\{a_n\}$ is Cauchy if for any $\varepsilon > 0\ \exists\ N \in \mathbb{N}$

s.t $|a_n - a_m| < \varepsilon \quad \forall n, m \geq N$



11/02/25 Lec-6

Theorem :- Let A be a set of real numbers. Then there is no onto map from A and P(A).

- $P(A) = \{B \subseteq \mathbb{R} : B \subseteq A\}$

Proof: Let $F : A \to P(A)$.

We define

$B = \{x : x \notin F(x)\}$

Note that $F(x) \subseteq A$

Ex → $F : A \to P(A)$, $F(x) = \{x\}$, $B = \phi$; $F : \mathbb{N} \to P(\mathbb{N})$, $F(x) = \{n+1\}$, $B = \mathbb{N}$

Then if F is an onto map, then $\exists y \in A$ such that $F(y) = B$.

The statement says $\exists$ any $y \in A$ st $F(y) = B$.

By Contradiction,

Let $\exists\ y_0$ such that $F(y_0) = B$.

Either $y_0 \in F(y_0) = B$

⇒ $y_0 \in B$

⇒ $y_0 \notin F(y_0)$

⇒ ⇐

$y_0 \notin F(y_0) = B$

⇒ $y_0 \notin B$

⇒ $y_0 \in F(y_0)$

⇒ ⇐ contradiction

Proved ⇒

$Car(\mathbb{N}) = \aleph_0$ (Celph ~~first~~ n naught)

$Car(\mathbb{R}) = \mathfrak{C}$ (Continuum)

Relations-

$\mathfrak{C} = 2^{\aleph_0}$

We can show that $\{0,1\}^{\mathbb{N}} \sim \mathbb{R}$

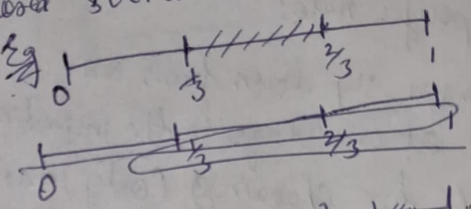
Cantor Set :- We denote Cantor Set by

$C \subseteq [0,1]$

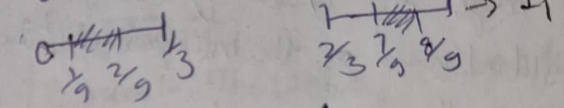
⇒ We construct the set by considering successfully removing "middle third" from the interval $[0, 1]$.

⇒ And continue the process inductively :- At the nth stage we construct I_n from I_{n-1} by removing 2^{n-1} disjoint, open middle third intervals from $I_n - 1$, each of length $\frac{1}{3^n}$.

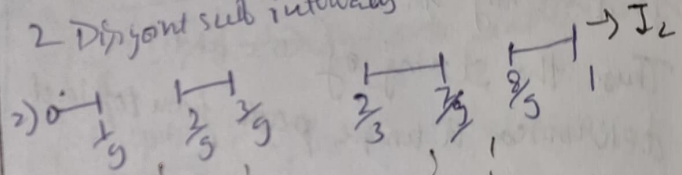
① Thus, I_n is the union of 2^n disjoint closed subinterval.



⇒



2 Disjoint sub intervals



4 Disjoint intervals

Means 2^n disjoint intervals.

We define the Cantor Set as

$C = \Delta = \bigcap_{n=1}^{\infty} I_n$

⇒ A Cantor set is an uncountable set.

- By nested interval theorem.

$\Delta \neq \phi$.

(Notice) : Δ (Cantor set) is atleast Countably infinite.

Claim : Δ is uncountable.

Proof :- First note that.

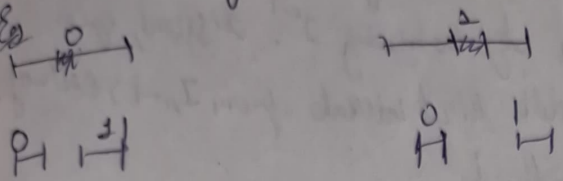
Each subinterval of I_{n-1} results

in 2 subintervals (disjoint) in I_n

We level these subintervals by 0 and 1.

0 → left subinterval.

1 → Right subinterval.



⇒ Imagine the subintervals are steps and a person stepping down with specific rule:

⇒ The person step down from one-subinterval at one stage to the immediate next stage by choosing (only) one of the subintervals of the interval he/she is presently in.

Thus the string of 0's and 1's determines a unique path from top level to bottom.

And each such "path" determined a unique set of nested intervals one from each level. whose intersection is a single point in Δ.

Conversely; each point $x \in \Delta$ can be achieved by following exactly one such path / steps.

If we consider a set as

$$S = \{0.a_1a_2a_3\cdots : a_n \in \{0,1\}\}$$

either 0 or 1

Clearly this set is uncountable ①

$f: S \to \Delta$ as

$f(s) = x$, where x is determined by the corresponding path given by the string in S.

By the defn of f we see that it is one-one (but onto as well).

Hence Δ is uncountable as S is. so (proved)

Theorem:- Δ is a "measure zero" set

Def (measure zero) = the length of it's complement is same as the length of the full set.

Proof: we show that the "length" of $[0,1]\setminus\Delta$ is exactly 1.

By induction, total length of 2^{n-1} disjoint subintervals in J_n = (the set of discarded set at n^{th} stage) is

$$\frac{2^{n-1}}{3^n}$$

∴ Therefore "length" $([0,1]\setminus\Delta)$

$$= \sum_{n=1}^{\infty} \frac{2^{n-1}}{3^n}$$

$$= \frac{1}{3}\sum_{n=1}^{\infty} \frac{2^{n-1}}{3^{n-1}}$$

$$= \frac{1}{3}\cdot\left(\frac{1}{1-\frac{2}{3}}\right) = 1$$

(proved)

Theorem:- If $x \in \Delta$, then $x = \sum_{n=1}^{\infty} \frac{a_n}{3^n}$, where a_n's are either 0 or 2.

Note:- $\frac{1}{3} = \sum_{n=2}^{\infty} \frac{2}{3^n}$

13/08/25 Lec-7

Properties:-

- Δ → Cantor set is uncountable.
- It is a "measure zero" set

↓

The complement having full length.

- If $x \in \Delta$; $x = \sum_{n=1}^{\infty} \frac{a_n}{3^n}$ where $a_n \in \{0, 2\}$

Cantor - Lebesgue - function:-

We take a function:-

$f: \Delta \to [0,1]$ such that

$f(x) = f\left(\sum_{n=1}^{\infty} \frac{2b_n}{3^n}\right)$ ← $b_n \in \{0,1\}$

$= \sum_{n=1}^{\infty} \frac{b_n}{2^n}$

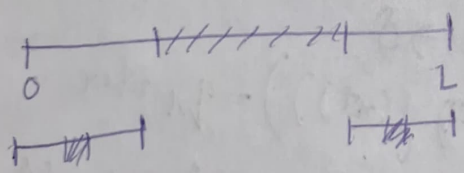
Remark: Given $x \in \Delta$ → seq of b_n's are fixed.

Properties:-

- $f(0) = 0$ and $f(1) = 1$

Result:-

f is a non-decreasing function, $\forall x, y \in \Delta$ and $x < y$ then $f(x) \le f(y)$.



Properties:-

i) f is not one-one function.

Claim:- $f\left(\frac{1}{3}\right) = f\left(\frac{2}{3}\right)$

$\frac{1}{3} = \sum_{n=2}^{\infty} \frac{2}{3^n}$, $\frac{2}{3} = \sum_{n=1}^{\infty} \frac{a_n}{3^n}$

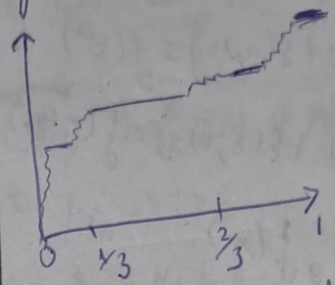
$n \in \{0, 2, 2, \dots\}$ $a_n \in \{2, 0, \dots\}$

$f\left(\frac{1}{3}\right) = \sum_{n=2}^{\infty} \frac{1}{2^n}$ $f\left(\frac{2}{3}\right) = \frac{1}{2}$

⇒ Define an extension of f to $[0,1]$.

$f(x) = \sup\{f(y) : y \in \Delta, y < x\}$

for $x \in [0,1]$



we get a monotonically increasing function

Property:-

Theorem:- The Cantor - Lebesgue fun $f: [0,1] \to [0,1]$ is continuous.

Property:- $f'(x) = 0$ 'almost everywhere' but the function increases from 0 to 1.

Monotone functions

Jump discontinuity:- A point c is call point of jump discontinuity if $f(c^+)$ and $f(c^-)$ exist.

Theorem:- Let $f: (a,b) \to \mathbb{R}$ be monotone, then f has only jump discontinuities.

Proof:- WLOG (without loss of generality) f is nondecreasing (If not take $-f$)

i) Let c be a point of discontinuity

As f is increasing $\{f(t) : t < c\}$ has an upper bound given by $f(c)$

Similarly $\{f(t) : c < t < b\}$ has a ~~increasing~~ lower bound given by $f(c)$.

To show

$\sup\{f(t) : t \in (a,b)\} = f(c^-)$

and $\inf\{f(t) : t \in (c,b)\} = f(c^+)$

$f(c^-) = \lim_{x \to c^-} f(x)$.

given $\varepsilon > 0$, $\exists\, t_0 \in (a,c)$ s.t

$\sup_{t \in (a,c)} f(t) - \varepsilon < f(t_0) \leq \sup_{t \in (a,c)}$

Denote $\delta = c - t_0$.

Then $\forall\, c - \delta < x < c$

$\sup f(t) - \varepsilon < f(t_0) < f(x)$
$t \in (a,c)$

$\Rightarrow f(c^-) = \sup f(t)$ $t \in (a,c)$

The other part (Exercise) proved.

Theorem:- If $f : (a,b) \to \mathbb{R}$ is monotone, then f has almost countably many points of discontinuity in (a,b)

proof:- The point of discontinuity must be of the form: jump discontinuity.

Assume:- f is increasing

Then if c is a point of discontinuity

$f(c^-) < f(c^+)$

Claim:- If c, d are two discontinuous points then $(f(c^-), f(c^+))$ and $(f(d^-), f(d^+))$ must be disjoint

To prove the claim, assume $c < d$

~~$\Rightarrow f(c) \neq f(d)$ (f is increasing)~~

$f(c^-) < f(c^+)$ and $f(d^-) < f(d^+)$

(It can never happen that $f(d)^- < f(c^+)$)

Implies

$\Rightarrow f(c^+) \leq f(d^-)$ so,

$(f(c^-), f(c^+))$ and $(f(d^-), f(d^+))$ are disjoint.

~~same real~~

~~Then we get a set of discontinuous open intervals of $\mathbb{R}$.~~

$S = \{x : x \text{ is point of discontinuity}\}$

$A = \{(f(x^-), f(x^+)), x \in S\}$

↳ must be a countable set.

$F : A \to \mathbb{Q}$

$F((f(x^-), f(x^+))) = q_x$ where

$q_x \in (f(x^-), f(x^+))$ and $q_x \in \mathbb{Q}$

Then A can not be an uncountable set.

Assignment-1

Q1 (if) Given set $A \subset \mathbb{R}$ that is bounded above. $\Rightarrow$ by least upper bound property $\exists s_0$ s.t $s_0 = \sup\{s : s \text{ is a upperbounds of } A\}$

$\exists \varepsilon > 0$ s.t $\forall a \in A \quad a \le s_0 - \varepsilon$.

$s_0 \ge a + \varepsilon \quad \forall a \ge a_0$ s.t

$a_0 + \varepsilon$ is an upperbound of A

s_0 is min $\Rightarrow\Leftarrow$

(only if) u is an upperbound of A.

$\varepsilon = u - s_0$, so lub of A of $\varepsilon > 0$,

$u \le s_0$

$a \in A_0 \cdot a > s_0 - \varepsilon$

~~$a > 2s_0 - u$~~

Q1 i) part

given $s_0 \to$ upper bound of A

$\sup A \le s_0$

$\varepsilon = s_0 - \sup A$, we take $\varepsilon > 0$

$\exists a_0$ s.t $a_0 > s_0 - \varepsilon$

$\cdot\ s_0 - s_0 + \sup A$

$\sup A < a_0$ $\Rightarrow\Leftarrow$ contradiction

$\Rightarrow s_0 = \sup A$.

Q2 If $\{x_n\} \xrightarrow{\text{converges}} x$.

Any subsequence $\{x_{n_a}\} \subseteq \{x_n\}$ also converges to x.

Converse :- Proof it by Contradiction

Assume $\{x_n\} \not\to x$

Q1 ii) part $\{x_n\}_n$. C A s.t

$\lim x_n = s_0$

Take $\varepsilon_n = \frac{1}{n}$, $\exists a_n$ it $a_n > s_0 - \frac{1}{n} \Leftrightarrow$

$|s_0 - a_n| = \frac{1}{n}$

$\{a_n\}$ Claim :- $\lim_{n\to\infty} a_n = \sup s_0$

Continue.....

for $\varepsilon > 0$, $\exists n_k > N \ \forall N \in \mathbb{N}$

s.t $|x_n - x| > \varepsilon$

Take $\{x_{n_k}\}$ in the subseq

given $\varepsilon > 0$, $\exists N_0$ s.t :

$|x_n - x| < \varepsilon \quad \forall n > N_0$] converges Defn

Take $\{x_n\} \subseteq \{x_k\}$ then

$\{x_{n_{k_j}} - x| > \varepsilon_0$

$\Rightarrow x_{n_{k_j}} \not\to x$

Q3 $\Leftrightarrow$ $a_n \ge 0$ and $\sum_{n=1}^{\infty} a_n < \infty$

i) Show that $\liminf_{n\to\infty} n a_n = 0$

ii) Let $t_n = \inf\{a_k : k \ge n\}$

$\alpha \ge 0 \quad \alpha = \liminf_{n\to\infty} n a_n$

if $\alpha > 0 \cdot \exists \delta$ s.t $0 < \alpha - \delta < 1$

$\varepsilon = \delta$

$\alpha - \delta < n a_n$ (By property lim inf)

$\forall n \ge N_0$

(M = limsup, m = liminf

$a_n < M + \varepsilon \ \forall n \ge N_0$)

$m - \varepsilon < a_n \ \forall n \ge N_0$

$\Rightarrow 0 < \frac{\alpha - \varepsilon}{n} < a_n$

$(\alpha - \varepsilon)\sum \frac{1}{n} \to$ diverges

$n a_n$

$\sum_{n.a_n}^{\infty} a_n \longrightarrow$ divergent.

ii) Example Let

$\limsup n a_n > 0$

$$a_n = \begin{cases} \frac{1}{2^k}, & n = 2^k \\ 0, & \text{elsewhere} \end{cases}$$

$\sum_{k=1}^{\infty} a_n < \infty$

$\limsup n a_n$

$= \inf \left[\sup_n \left[\delta_k \quad k > n \right] \right]$

$\inf = 1$

(5) (a) $x_n = \left(\cos \frac{n\pi}{4}\right)^{-1}$

$x_1 = \sqrt{2}$

$x_2 = 0$

$x_3 = -\sqrt{2}$

$x_4 = -1$

$\liminf x_n$

$\limsup x_n$

(b) (c) $a_n = \begin{cases} \frac{n}{n+1} & n \text{ odd} \\ \frac{1}{n+1} & n \text{ even} \end{cases}$

Oscillatory

$\liminf a_n = 0$

$\limsup a_n = 1$